

Member ID: _____

Time: _____

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LINUX OPERATING SYSTEM FUNDAMENTALS (350)

STATE 2024

Total Points: (50 @ 2 points each) _____(100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

Directions: Identify the letter of the choice that *best* completes the statement or answers the question.

1. Which of the following commands is used to display the contents of a text file?
 - A. cat
 - B. touch
 - C. echo
 - D. mv
2. Which service is responsible for managing user authentication?
 - A. SSH
 - B. DHCP
 - C. LDAP
 - D. FTP
3. What does the 'SSH' protocol stand for?
 - A. Secure Socket Host
 - B. Simple Secure Hosting
 - C. Secure Shell
 - D. Server System Hosting
4. Which command is used to create a backup of a directory and its contents?
 - A. copy
 - B. mv
 - C. tar
 - D. cp
5. What is the purpose of the 'sudo' command?
 - A. Switch to a different user account
 - B. Execute commands with superuser privileges
 - C. Create a new user account
 - D. Display system logs
6. Which command is used to view and manipulate system logs?
 - A. syslog
 - B. logview
 - C. journalctl
 - D. logwatch
7. What does the acronym 'SELinux' stand for in the context of Linux security?
 - A. Secure Environment Linux
 - B. Security-Enhanced Linux
 - C. System Event Logging Linux
 - D. Service Execution Linux

8. In Linux, which command is used to change the ownership of a file or directory?
 - A. own
 - B. chown
 - C. chmod
 - D. group
9. Which protocol is commonly used for remote file transfer?
 - A. FTP
 - B. SSH
 - C. HTTP
 - D. Telnet
10. In Linux, what does the 'systemd' init system primarily handle?
 - A. User authentication
 - B. System logging
 - C. Service management and initialization
 - D. Network configuration
11. Which directory contains system-specific startup scripts and initialization files?
 - A. /bin
 - B. /sbin
 - C. /lib
 - D. /etc/init.d
12. What is the purpose of the 'passwd' command with the '-l' option?
 - A. Lock a user account
 - B. List user accounts
 - C. Change the root password
 - D. Log out of the current session
13. Which command is used to display the contents of a compressed file?
 - A. unzip
 - B. untar
 - C. decompress
 - D. zcat
14. What is the purpose of the 'netstat' command?
 - A. List all installed packages
 - B. Display system uptime
 - C. Display network statistics
 - D. List directory contents
15. What is the purpose of the 'ping' command?
 - A. List network interfaces
 - B. Display system load average
 - C. Test network connectivity
 - D. Display system time

16. Which utility is used to create and manipulate disk partitions?
 - A. df
 - B. fdisk
 - C. du
 - D. mkfs
17. Which directory contains user-specific configuration files and settings?
 - A. /bin
 - B. /usr
 - C. /home
 - D. /etc
18. What does the 'SSH' protocol use for secure authentication?
 - A. FTP
 - B. HTTP
 - C. Public-key cryptography
 - D. Telnet
19. What does the acronym 'UDP' stand for in the context of networking?
 - A. Universal Data Protocol
 - B. Uniform Datagram Protocol
 - C. User Datagram Protocol
 - D. Unified Data Processing
20. In Linux, what is the primary purpose of the 'sar' command?
 - A. Create compressed archives
 - B. Monitor system performance
 - C. Configure network interfaces
 - D. Manage user accounts
21. What is the purpose of the 'nmap' command?
 - A. Create network maps
 - B. Display system logs
 - C. Perform network scanning and mapping
 - D. Manage network interfaces
22. Which file is used to specify DNS servers?
 - A. /etc/hosts
 - B. /etc/dns.conf
 - C. /etc/resolv.conf
 - D. /etc/network
23. Which command is used to create a symbolic link?
 - A. symlink
 - B. link
 - C. ln -s
 - D. slink

24. Which file is used to define custom user aliases?
- A. /etc/passwd
 - B. ~/.aliases
 - C. ~/.bashrc
 - D. /etc/aliases
25. Which command is used to display the routing table?
- A. route
 - B. netstat
 - C. ifconfig
 - D. ipconfig
26. You are troubleshooting a network issue and want to check the network connectivity between your machine and a remote server. Which command can help you perform this test?
- A. ping
 - B. netstat
 - C. traceroute
 - D. ifconfig
27. Which command can you use if you need to generate a secure random password of 12 characters in length in the terminal?
- A. pwgen 12
 - B. openssl rand -base64 12
 - C. randompass -l 12
 - D. passgen -n 12
28. You want to view detailed information about a specific network interface, including its IP address and MAC address. Which command should you use?
- A. ifconfig
 - B. ipconfig
 - C. netstat
 - D. ifstatus
29. You need to change the default runlevel of a Linux system to boot into text mode instead of the graphical interface. Which runlevel should you set in the /etc/inittab file?
- A. 3
 - B. 5
 - C. S
 - D. G
30. You are tasked with changing the hostname of a Linux server. Which file should you edit to make this change persist after a reboot?
- A. /etc/hostname
 - B. /etc/hosts
 - C. /etc/network/interfaces
 - D. /etc/sysconfig/network

31. You are troubleshooting a network issue, and you want to check the DNS resolution for a specific domain. Which command should you use to test DNS resolution?
- A. ping
 - B. nslookup
 - C. traceroute
 - D. dig
32. You need to schedule a task to run every Monday at 2:30 AM. Which entry should you add to your crontab file?
- A. 30 2 * * 1 command
 - B. 30 2 * * 0 command
 - C. 30 2 * * 7 command
 - D. 30 2 * * 2 command
33. You need to find out which Linux distribution and version you are using from the command line. Which command should you use?
- A. version
 - B. uname -r
 - C. lsb_release -a
 - D. cat /etc/os-release
34. You need to check the open ports on a remote server. Which command should you use for port scanning?
- A. ping
 - B. netstat
 - C. nmap
 - D. telnet
35. You are configuring a firewall rule to allow incoming FTP traffic. Which protocol should you specify in the rule?
- A. TCP
 - B. UDP
 - C. SFTP
 - D. FTP
36. You want to monitor real-time CPU and memory usage of all running processes on your Linux system. Which command should you use?
- A. top
 - B. ps -e
 - C. htop
 - D. df -h
37. You are configuring a firewall rule to allow incoming DNS traffic. Which port should you specify in the rule?
- A. 22
 - B. 53
 - C. 80
 - D. 443

38. You are configuring a firewall rule to block outgoing web traffic (HTTP and HTTPS). Which ports should you block?
- A. 80 and 443
 - B. 22 and 53
 - C. 8080 and 8443
 - D. 25 and 110
39. You want to securely copy a file from your local machine to a remote server using SSH. Which command should you use?
- A. scp
 - B. ftp
 - C. rsync
 - D. telnet
40. You suspect that a service is not listening on the expected port. Which command can you use to check which processes are listening on specific ports?
- A. netstat -l
 - B. ps -ef
 - C. lsof -i :port
 - D. chkconfig --list
41. You need to monitor the network traffic to and from a specific IP address on your Linux server. Which command can help you achieve this?
- A. tcpdump -i eth0 host specific_ip
 - B. netstat -n specific_ip
 - C. wireshark specific_ip
 - D. nmap -sP specific_ip
42. You want to find all files on your system that were modified within the last 7 days. Which command should you use?
- A. find / -type f -mtime -7
 - B. grep -r "modified: 7 days ago"
 - C. ls -lt | grep "7 days"
 - D. locate -m 7
43. You need to find all running processes that contain the word "apache" in their command names. Which command should you use?
- A. ps aux | grep apache
 - B. pgrep -f apache
 - C. ps -e | grep apache
 - D. top -c apache

44. You want to list all active network connections on your Linux system, including information about the state of each connection. Which command should you use?
- A. netstat -l
 - B. lsof -i
 - C. ss -tuln
 - D. ifconfig -a
45. Which Linux distribution is known for its use in embedded systems and is designed for simplicity and ease of use?
- A. CentOS
 - B. Debian
 - C. Ubuntu
 - D. Raspberry Pi OS
46. You want to limit the number of login attempts for a specific user. Which configuration file should you modify?
- A. /etc/passwd
 - B. /etc/security/limits.conf
 - C. /etc/login.defs
 - D. /etc/pam.d/common-auth
47. You want to run a Linux command in the background and ensure that it continues running even after you log out of the terminal. Which command should you use?
- A. bg
 - B. nohup
 - C. daemon
 - D. disown
48. You need to find all files containing a specific text string within a directory and its subdirectories. Which command should you use?
- A. grep -r "text" directory
 - B. find directory -type f -exec grep "text" {} +
 - C. ls -l directory | grep "text"
 - D. cat directory | grep -r "text"
49. You need to change the default shell for a user to /bin/zsh. Which command should you use to modify the user's shell?
- A. chsh -s /bin/zsh username
 - B. usermod -s /bin/zsh username
 - C. passwd -s /bin/zsh username
 - D. setshell /bin/zsh username
50. You want to display the contents of a file while continuously updating the view as new lines are added to the file. Which command should you use?
- A. cat -f filename
 - B. more +F filename
 - C. tail -f filename

D. `watch -n 1 cat filename`